

Title: AI in Smart Resume Screening and Job Matching System

Domain/Field:

- Artificial Intelligence (AI)
- Natural Language Processing (NLP)

Problem Statement:

Traditional resume screening is:

- Time-consuming
- Biased
- Inefficient

The paper proposes an AI-based automated system to:

- Parse resumes
- Match them with job descriptions
- Rank candidates fairly and efficiently

Summary

This paper introduces a smart system that helps companies automatically select the best candidates from resumes.

In simple terms:

- Instead of humans reading hundreds of resumes,
- The system reads and understands resumes using AI.

What it does:

1. Extracts important details (skills, education, experience)
2. Compares them with job requirements
3. Gives each candidate a score
4. Ranks candidates based on suitability

Methodology (Step-by-Step)

Step 1: Resume Input

- Upload resumes (PDF/DOCX)

Step 2: Text Extraction & Preprocessing

- Clean text (remove noise, normalize format)

Step 3: Information Extraction

- Use spaCy & NLTK
- Extract:
 - Skills
 - Experience
 - Education

Step 4: Job Description Processing

- Convert job description into vectors using:
 - TF-IDF
 - Word2Vec

Step 5: Similarity Calculation

- Use Cosine Similarity:
 - Measures how similar resume is to job description

Step6: Scoring Model Weights:

- Skills → 40%
- Experience → 30%
- Education → 20%
- Certifications → 10%

Step 7: Ranking

- Generate final compatibility score
- Rank candidates accordingly

Step 8: Bias Reduction

- Remove personal details (name, gender, etc.)

Step 9: Explainability

- Show:
 - Why candidate got that score

- Key contributing features

Tech Stack:

- Backend: Python, Flask/Django
- Frontend: React.js
- **AI/NLP:**
 - spaCy
 - BERT
- Database: MySQL
- **Tools:**
 - Scikit-learn
 - Streamlit